Stanford Andrew Ng. Least Squares

(Youtube — Coursera)

exemplu $A = \begin{pmatrix} 2 & 6 \\ 1 & -3 \\ 2 & 0 \end{pmatrix}$ $b = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

$Ax = b$ $x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

$\Rightarrow \begin{cases} 2x_1 + 6x_2 = 1 \\ x_1 - 3x_2 = 0 \\ 2x_1 = 0 \end{cases}$ ✗ nu.

$x_2 = 0$

$x_1 = 0$

incompatibil.

$Ax^* = \text{pr}_{\text{Im}(A)} b$

$\text{Im}(A) = \text{Sp}\left\{ \underbrace{\begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}}_{v_1}, \underbrace{\begin{pmatrix} 6 \\ -3 \\ 0 \end{pmatrix}}_{v_2} \right\}$ GS

$\{v_1, v_2\}$ bază în $\text{Im}(A) \longrightarrow g_1 = \frac{1}{3}\begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}$,

$g_2 = \frac{1}{3}\begin{pmatrix} 2 \\ -2 \\ -1 \end{pmatrix}$

bază ortonormată în $\text{Im}(A)$

$Q = (q_1 / q_2)$

$$\Rightarrow \underset{\mathrm{Im}(A)}{\mathrm{pr}}\, b = \underline{QQ^t b} = \frac{1}{3}\begin{pmatrix} 2 & 2 \\ 1 & -2 \\ 2 & -1 \end{pmatrix} \frac{1}{3}\begin{pmatrix} 2 & 1 & 2 \\ 2 & -2 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$= \frac{1}{9}\begin{pmatrix} 8 \\ -2 \\ 2 \end{pmatrix}$$

$$\rightarrow = \underset{q_1}{\mathrm{pr}}\, b + \underset{q_2}{\mathrm{pr}}\, b = \underline{\langle b, q_1 \rangle q_1 + \langle b, q_2 \rangle q_2}$$

$$\begin{cases} 2x_1^* + 6x_2^* = 8/9 \\ x_1^* - 3x_2^* = -2/9 \\ 2x_1^* = 2/9 \end{cases} \Rightarrow \begin{array}{l} x_2^* = \frac{1}{9} \\ x_1^* = \frac{1}{9} \end{array} \quad \checkmark$$

pseudo-soluția

Sistemul inițial incompat.

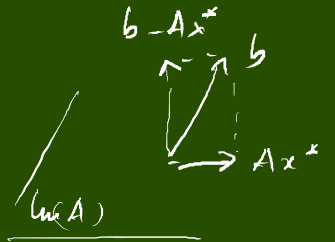
$$Ax^* = \underset{\mathrm{Im}(A)}{\mathrm{pr}}\, b$$

$\mathrm{rang}\, A = n \leq m \Rightarrow \exists\, Q, R.\ A = \underline{QR}$

$$QRx^* = QQ^t b \quad / \cdot Q^t$$

$$\Rightarrow Rx^* = Q^t b \quad \Rightarrow \underline{x^* = R^{-1} Q^t b}$$

$$Ax^* = \text{pr}_{\text{Im}(A)} b \iff \underline{A^t A x^* = A^t b}$$



$$Ax^* = \text{pr}_{\text{Im}(A)} b \iff$$

$$b - Ax^* \in \text{Im}(A)^{\perp} = \text{Ker}(A^t)$$

$$\iff A^t (b - Ax^*) = 0$$

$$\iff A^t A x^* = A^t b$$

$A^t A x^* = A^t b$ sol. unica $\iff$

$(n,n)(n,n)$ $\quad A^t A$ invertibile

(n,n)



$\operatorname{rang} A = n \leq m$

$A^t A$ inversabilă $\Leftrightarrow$ $\ker(A^t A) = \{0\}$

$\ker(A^t A) = ?$

$x \in \ker(A) \Rightarrow Ax = 0 \quad / \cdot A^t \Rightarrow$

$\Rightarrow A^t A x = 0 \Rightarrow x \in \ker(A^t A)$

$\ker(A) \subseteq \ker(A^t A)$

$\supseteq$

$x \in \ker(A^t A) \Rightarrow A^t \underset{(n,1)}{A x} = 0 \quad / \cdot x^t$

$(?, n)(n, m)\,(m, n)$

$\underbrace{x^t A^t}_{(Ax)^t} A x = 0$

$$\underbrace{(Ax)^t \cdot (Ax)}_{\langle Ax, Ax \rangle} = 0 \Rightarrow Ax = 0$$

$$x \in \ker(A)$$

$$\langle Ax, Ax \rangle = \|Ax\|^2$$

$$\Rightarrow \ker(A) = \ker(A^tA)$$

$$\dim_{\mathbb{R}} \ker(A) = n - \operatorname{rang}(A)$$

$\Rightarrow A^tA$ invertible $\Leftrightarrow \operatorname{rang}(A) = n$.

$U \subseteq \mathbb{R}^m \quad b \in \mathbb{R}^m$

$U \longrightarrow$ base $\{v_1 \dots v_n\}$

$$A = (v_1 | \dots | v_n)$$

$U = \operatorname{Im}(A)$ $\quad p_U b = A \cdot (A^tA)^{-1} A^t b$

$A^\dagger$ $(A^tA)^{-1}A^t$ pseudoinversa

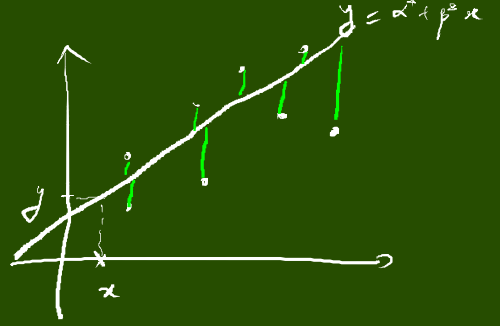
dagger Moore - Penrose

tema

$$\begin{cases} 2x_1 + 6x_2 = 1 \\ x_1 - 3x_2 = 0 \\ 2x_1 = 0 \end{cases}$$

A, $x^* = (A^tA)^{-1}A^tb$

$= \dots = \begin{pmatrix} 1/9 \\ 1/9 \end{pmatrix}$



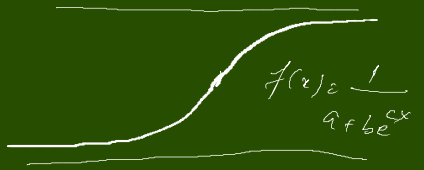
scatter plot

$y = \alpha + \beta x + \gamma x^2 \longrightarrow \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}$

$$A = \begin{pmatrix} 1 & x_1 & x_1^2 \\ \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 \end{pmatrix}$$

$$y = \alpha e^{\beta x}$$

$$\ln y = \ln \alpha + \beta x$$



$$\beta^* = 0,00001$$

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1) $f(x_1 \dots x_n)$

$$\frac{\partial f}{\partial x_i}(x_1 \dots x_n) = 0 \quad \forall i = \overline{1,n}$$

$$\left(\frac{\partial^2 f}{\partial x_i \partial x_j} \right)_{i,j = \overline{1,n}} = H_f \quad \text{Hessiana}$$

$f \in C^2 \Rightarrow H_f$ simetrică

- maxim: $\forall \lambda < 0$
- minim: $\forall \lambda > 0$

2) Statistică, ML

$x_1 \dots x_n$ ← media $\mu = \frac{\sum x_i}{n}$

dispersione $\sigma^2 = \sum \frac{(x_i - \mu)^2}{n-1}$

$X = (x_1 | \dots | x_n) \in M_{n,n}(\mathbb{R})$ ⟶ covariante

$\frac{1}{n-1}(X-\mu)(X-\mu)^t$

simmetrica

$A \in M_n(\mathbb{R})$ simmetrica $A = A^t$

$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ $X^2 + 1$ ⟨ i, $-i$

$A = A^t \Rightarrow \sigma(A) \subseteq \mathbb{R}$

pp. $\exists \lambda \in \mathbb{C}$ $\lambda \in \sigma(A)$ tale che $v \in \mathbb{C}^n$, $v \neq 0$
vedo proprio

$Av = \lambda v$

$\bar{A}\bar{v} = \bar{\lambda}\bar{v} \Rightarrow A\bar{v} = \bar{\lambda}\bar{v}$

$A \in \mathcal{U}_n(\mathbb{R})$ $\quad \bar{A} = A$

$$\lambda \bar{v}^t v = \bar{v}^t A v = \underbrace{\bar{v}^t A^t v}_{(A\bar{v})^t} = (\bar{\lambda}\bar{v})^t v =$$

$$= \bar{\lambda}\, \bar{v}^t v$$

$\underbrace{(1,n)(n,1)}_{\in \mathbb{C}}$

$$v = \begin{pmatrix} z_1 \\ \vdots \\ z_n \end{pmatrix} \neq 0 \qquad \bar{v}^t v = (\bar{z}_1 \dots \bar{z}_n)\begin{pmatrix} z_1 \\ \vdots \\ z_n \end{pmatrix} =$$

$$= \bar{z}_1 z_1 + \dots + \bar{z}_n z_n$$

$$= |z_1|^2 + \dots + |z_n|^2 > 0$$

$$\lambda(\bar{v}^t v) = \bar{\lambda}(\bar{v}^t v)$$

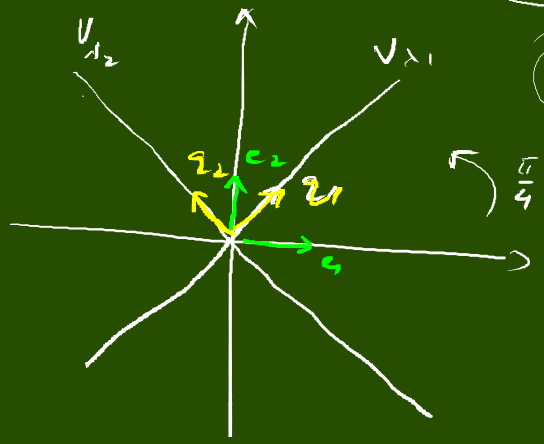
$$\underbrace{\bar{v}^t v}_{\neq 0}(\lambda - \bar{\lambda}) = 0 \quad \rightsquigarrow \quad \lambda = \bar{\lambda} \Rightarrow \lambda \in \mathbb{R}$$

esempio $\qquad A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \qquad A = A^{-t}$

$$P_A(x) = x^2 - 4x + 3 \begin{cases} \lambda_1 = 3 \\ \lambda_2 = 1 \end{cases}$$

$\lambda_1 = 3$ $\quad \begin{cases} 3x + y = 3x \\ x + 2y = 3y \end{cases} \Rightarrow x = y$

$$V_{\lambda_1} = \left\{ \begin{pmatrix} x \\ x \end{pmatrix} \mid x \in \mathbb{R} \right\} = Sp\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\} \quad q_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$



$A = Q \cdot D \cdot Q^t$

$\frac{\pi}{4}$

$P \cdot D \cdot P^{-1} = Q \cdot D \cdot Q^t$

$\lambda_2 = 1$ $\quad \begin{cases} x + y = x \\ x + 2y = y \end{cases} \quad x = -y$

$$V_{\lambda_2} = \left\{ \begin{pmatrix} -y \\ y \end{pmatrix} \mid y \in \mathbb{R} \right\} = Sp\left\{ \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\} \quad q_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$A = A^t$, $\lambda_1, \lambda_2 \in \sigma(A)$ $\lambda_1 \neq \lambda_2$

$v_1 \quad v_2$

$\Rightarrow v_1 \perp v_2$

$\lambda_1 \langle v_1, v_2 \rangle = \langle \lambda_1 v_1, v_2 \rangle = \langle A v_1, v_2 \rangle =$

$= (A v_1)^t v_2 = v_1^t A^t v_2 = v_1^t A v_2 = v_1^t \lambda_2 v_2 =$

$= \lambda_2 \langle v_1, v_2 \rangle$

$(\lambda_1 - \lambda_2) \langle v_1, v_2 \rangle = 0 \Rightarrow v_1 \perp v_2$

$\neq 0$

esempio (cont.) $A = Q \cdot D \cdot Q^t$

$Q = (q_1 | q_2) = \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}$

matrice ortogonale $Q^t Q = I$

$Q^{-1} = Q^t$

$\downarrow$

$(\det Q)^2 = 1$

$D = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$

$n = 2 \quad \begin{cases} 1 \\ -1 \end{cases}$ rotative

Exemplu. $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$

$\det(A) = 0 \Rightarrow \lambda = 0$ val. pr. pt. A

$\lambda = 3$ val. pr. pt. A.

$$P_A(x) = \begin{vmatrix} 1-x & 1 & 1 \\ 1 & 1-x & 1 \\ 1 & 1 & 1-x \end{vmatrix} = (3-x)\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1-x & 1 \\ 1 & 1 & 1-x \end{vmatrix} =$$

$$= (3-x)\begin{vmatrix} 1 & 1 & 1 \\ 0 & -x & 0 \\ 0 & 0 & -x \end{vmatrix} = (3-x)(-x)^2 = x^2(3-x)$$

$\lambda_1 = 3 \quad a(\lambda_1) = 1$

$\lambda_2 = 0 \quad a(\lambda_2) = 2$

$\lambda_1 = 3$

$$\begin{cases} x+y+z = 3x \\ x+y+z = 3y \\ x+y+z = 3z \end{cases} \Rightarrow x = y = z$$

$$V_{\lambda_1} = \left\{ \begin{pmatrix} x \\ x \\ x \end{pmatrix} \mid x \in \mathbb{R} \right\} = \mathrm{Sp}\left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$$

$$\Rightarrow \left\{ q_1 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\} \text{ base ortonormata di } V_{\lambda_1}$$



$\lambda = 0 \quad V_{\lambda_2} = V_{\lambda_1}^{\perp}$

$$\begin{cases} x+y+z = 0 \\ x+y+z = 0 \\ x+y+z = 0 \end{cases}$$

$$V_{\lambda_2} = \left\{ \begin{pmatrix} x \\ y \\ -x-y \end{pmatrix} \mid x, y \in \mathbb{R} \right\} = \mathrm{Sp}\left\{ \underbrace{\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}}_{v_2}, \underbrace{\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}}_{v_3} \right\}$$

GS $\longrightarrow$

$$u_2 = v_2 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

$$u_3 = v_3 - \mathrm{pr}_{u_2} v_3 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} -1/2 \\ 1 \\ -1/2 \end{pmatrix}$$

$$q_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

$$q_3 = \sqrt{\frac{2}{3}} \begin{pmatrix} -1/2 \\ 1 \\ -1/2 \end{pmatrix} = \frac{1}{\sqrt{6}} \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$$

$Q = (q_1 | q_2 | q_3)$ matrice ortogonală

$D = \begin{pmatrix} 3 & & \\ & 0 & \\ & & 0 \end{pmatrix}$

$Q Q^t \cdot Q Q^t = Q Q^t$

$(n,1)(1,n)$

(n,n)

$\Rightarrow \underline{A} = Q \cdot D \cdot Q^t$

$\underline{q_1 q_1^t} + \dots + \underline{q_n q_n^t} = I_n$

$\operatorname{pr}_{q_1} x + \dots + \operatorname{pr}_{q_n} x = x$ $\quad Sp\{q_1 \dots q_n\} = \mathbb{R}^n$

bază ortonormată $\mathbb{R}^n$.

Temă $\quad A \in M_n(\mathbb{R})$, $A = A^t$ $\quad A = (a_{ij})_{i,j}$

$(1,n)(n,n)(n,1)$

$f(x) = x^t A x \qquad x \in \mathbb{R}^n$

$f : \mathbb{R}^n \longrightarrow \mathbb{R}$

$f\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \sum a_{ij} x_i x_j$ cont.

1) $\max_{x \in \mathbb{R}^4, \|x\|=1} f(x) = ?$ $\quad x_M = ?$

2) $\min_{x \in \mathbb{R}^4, \|x\|=1} f(x) = ?$ $\quad x_m = ?$

3) $\max_{x \in \mathbb{R}^4, \|x\|=1, x \perp \xi_1} f(x)$

$A: \lambda_1 \geq \dots \geq \lambda_4$

$\{\xi_1, \dots, \xi_4\}$ bază ortonormată $\mathbb{R}^4$
